

Approximating Shortest Path Distance Filtering by Random Walks

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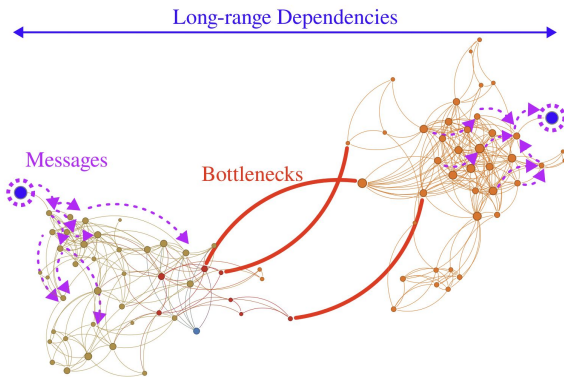
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Background

Limitations of Existing Approaches on Graph Representation Learning

Message Passing Neural Networks

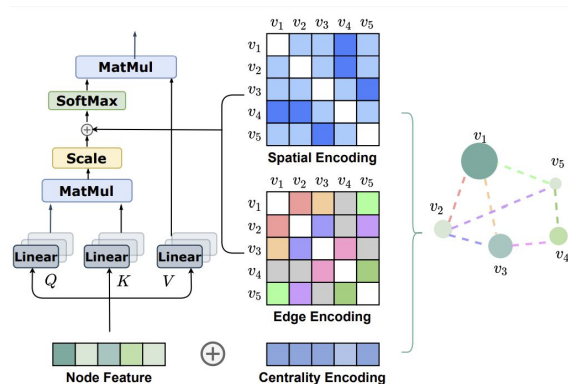
- Over-smoothing
- Over-squashing



On the Trade-off between Over-smoothing and Over-squashing in Deep Graph Neural Networks (CIKM 2023)

Graph Transformers

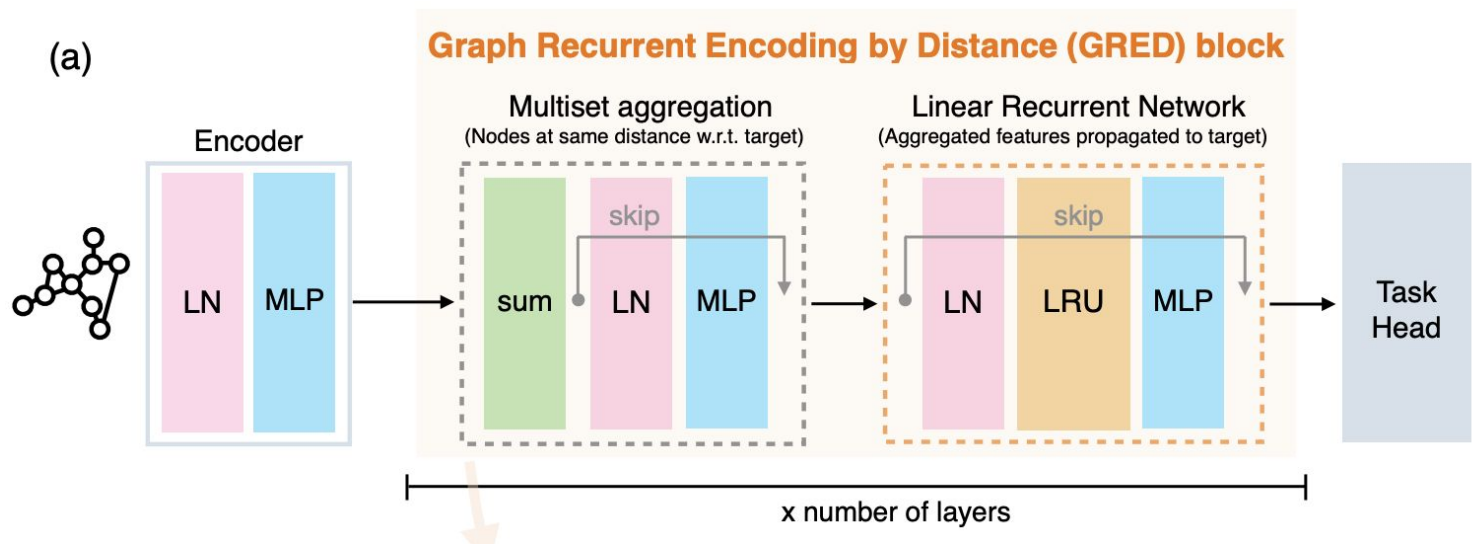
- Quadratic computational cost
- Needs ad-hoc positional encoding



Do Transformers Really Perform Bad for Graph Representation? (NeurIPS 2021)

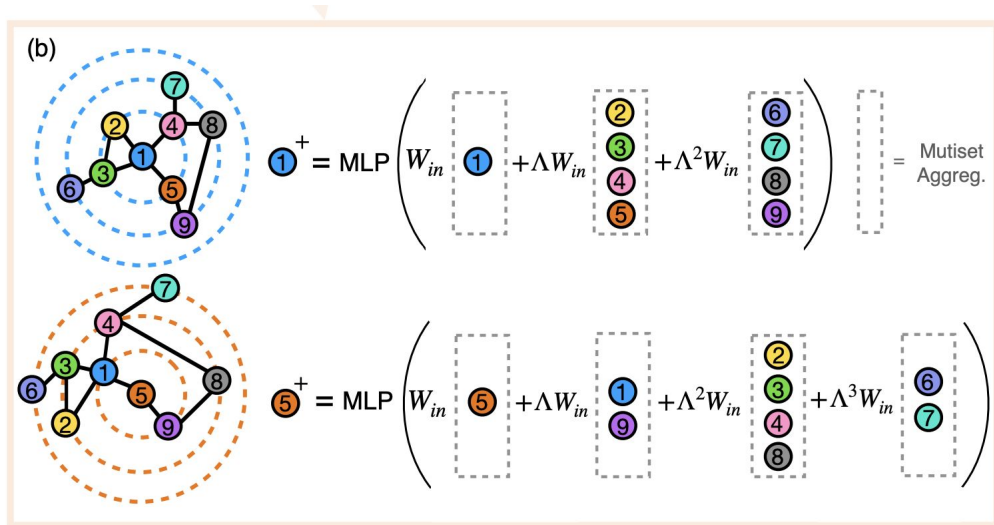
Preliminaries: GRED

GRED: Recurrent Distance Filtering for Graph Representation Learning [ICML 2024]



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$$\mathbf{x}_{v,k}^{(\ell)} = \text{MLP}_2 \left(\sum_{u \in \mathcal{N}_k(v)} \text{MLP}_1 \left(\mathbf{h}_u^{(\ell-1)} \right) \right) \in \mathbb{R}^d.$$

$$\mathbf{s}_{v,k}^{(\ell)} = \Lambda \mathbf{s}_{v,k-1}^{(\ell)} + \mathbf{W}_{in} \mathbf{x}_{v,K-k}^{(\ell)}$$

$$\mathbf{h}_v^{(\ell)} = \text{MLP}_3 \left(\Re \left[\mathbf{W}_{out} \mathbf{s}_{v,K}^{(\ell)} \right] \right)$$

Limitation of GRED

- While LRU-based model demonstrates **improvement on the LRGB** dataset that requires learning long-range dependency for the prediction.
- The model **failed to attain improvement** on five widely used general graph datasets.

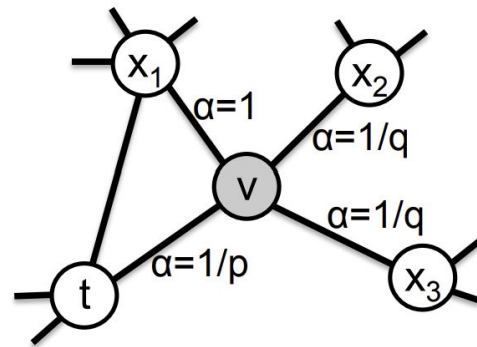
Model	MNIST	CIFAR10	PATTERN	CLUSTER	Model	Test MAE ↓
GCN (Kipf & Welling, 2017)	90.705±0.218	55.710±0.381	85.614±0.032	69.026±1.372	GCN (Kipf & Welling, 2017)	0.278±0.003
GAT (Veličković et al., 2018)	95.535±0.205	64.223±0.455	78.271±0.186	70.587±0.447	GAT (Veličković et al., 2018)	0.384±0.007
GIN (Xu et al., 2019)	96.485±0.252	55.255±1.527	85.590±0.011	64.716±1.553	GIN (Xu et al., 2019)	0.387±0.015
GatedGCN (Bresson & Laurent, 2017)	97.340±0.143	67.312±0.311	85.568±0.088	73.840±0.326	GatedGCN (Bresson & Laurent, 2017)	0.282±0.015
EGT (Hussain et al., 2022)	98.173±0.087	68.702±0.409	86.821±0.020	79.232±0.348	PNA (Corso et al., 2020)	0.188±0.004
SAN (Kreuzer et al., 2021)	-	-	86.581±0.037	76.691±0.65	KP-GIN (Feng et al., 2022)	0.093±0.007
SAT (Chen et al., 2022)	-	-	86.848±0.037	77.856±0.104	PathNN (Michel et al., 2023)	0.090±0.004
GPS (Rampášek et al., 2022)	98.051±0.126	72.298±0.356	86.685±0.059	78.016±0.180	SAN (Kreuzer et al., 2021)	0.139±0.006
Graph MLP-Mixer (He et al., 2023)	98.320±0.040	73.960±0.330	-	-	Graphormer (Ying et al., 2021)	0.122±0.006
GRIT (Ma et al., 2023)	98.108±0.111	76.468±0.881	87.196±0.076	80.026±0.277	K-subgraph SAT (Chen et al., 2022)	0.094±0.008
GRED (Ours)	98.383±0.012	76.853±0.185	86.759±0.020	78.495±0.103	GPS (Rampášek et al., 2022)	0.070±0.004
					Graph MLP-Mixer (He et al., 2023)	0.073±0.001
					GRIT (Ma et al., 2023)	0.059±0.002
					GRED (Ours)	0.077±0.002

Motivation

- Multiset Aggregation of GRED
 - GRED computes a target node's multiset based on the shortest path distance
 - Multiset aggregation is performed for each hop within a single layer
- Pros
 - Effectively Mitigates the **over-squashing problem**
- Cons
 - Fails to address the **over-smoothing problem** due to its **multiset aggregation strategy**
 - Performs sum aggregation on the representations of each hop distance **within a single layer**

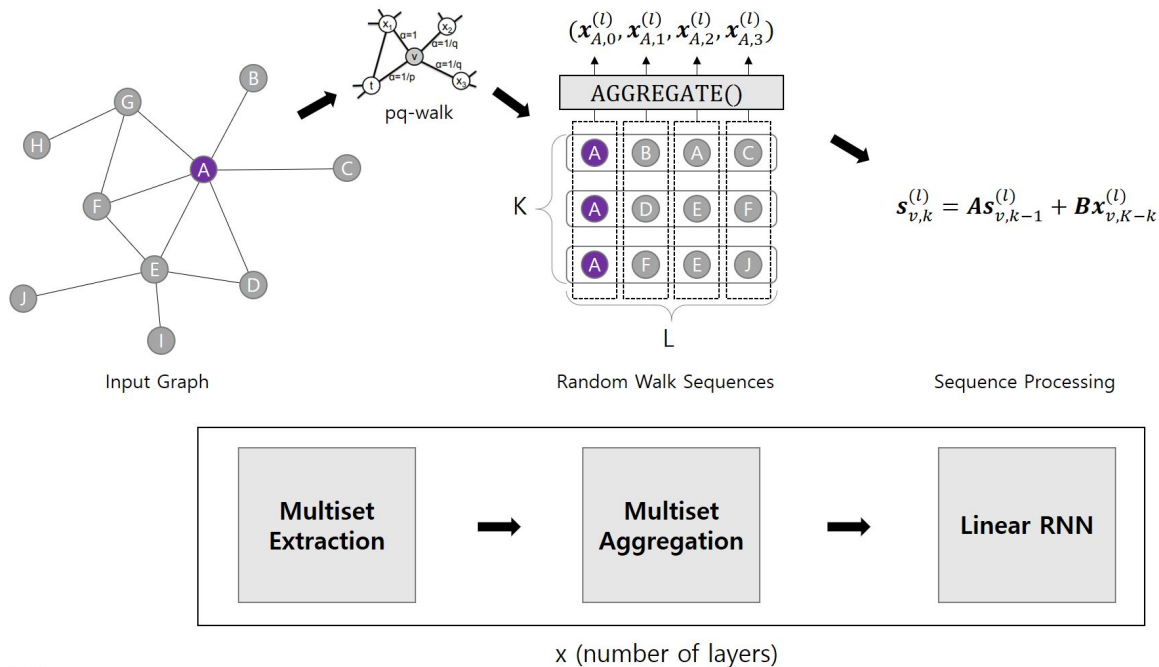
Method: Original Plan

- Replace the shortest path distance with **random walks**
 - Approximates shortest path distance by **utilizing position of each node** included in random walk sequences.
 - Reflect the **graph structure** using **p-q random walk** inspired by node2vec.
- Learn the **graph-specific** p, q value
 - Sample p-q random walk sequence on each epoch
 - Learn p,q value by applying straight-through trick
- Problem
 - Too **time-consuming** to conduct random walk on each epoch



node2vec: Scalable Feature Learning for Networks (KDD 2016)

Method



Conduct multiset aggregation by **position of nodes** in K number of random walk sequences.

Fine-tune the model with **different p and q value** for the random walk

Experiments: Setup

- **Dataset**

- **Super pixel based graph:** MNIST
- **Stochastic Block Model-based graph:** PATTERN, CLUSTER
- **Molecular graph:** ZINC(12K)

- **Baselines**

- **MPNNs:** GCN, GAT, GIN, GatedGCN
- **Multi-hop GNNs:** KP-GIN, PathNN
- **Graph Transformers:** SAN, SAT, GPS, Graph MLP-Mixer, GRIT
- **Linear Recurrent Unit-based GNN:** GRED

Dataset	#Graphs	#Classes	Avg. Nodes	Avg. Edges	Node feat.	Edge feat.	Task
MNIST	70,000	10	70.57	564.53	Pixel+Coord (3)	Node Dist (1)	Graph Classification
PATTERN	14,000	2	117.47	4,749.15	Node Attr (3)	N. A.	Node Classification
CLUSTER	12,000	6	117.20	4,301.72	Node Attr (7)	N. A.	Node Classification
ZINC	12,000	-	23.16	49.83	Atom Type (28)	Bond Type (4)	Graph Regression

Experiments: Main Result

Category	Model	Dataset			
		MNIST Accuracy(↑)	PATTERN Accuracy(↑)	CLUSTER Accuracy(↑)	ZINC 12K MAE(↓)
MPNN	GCN (Kipf & Welling, 2017)	90.705	85.614	69.026	0.278
	GAT (Velickovic et al., 2018)	95.535	78.271	70.587	0.384
	GIN (Xu et al., 2019)	96.485	85.590	64.716	0.387
	GatedGCN (Bresson & Laurent, 2017)	97.340	85.568	73.840	0.282
Multi-hop GNN	KP-GIN (Feng et al., 2022)	-	-	-	0.093
	PathNN (Michel et al., 2023)	-	-	-	0.090
Graph-Transformer	SAN (Kreuzer et al., 2021)	-	-	-	0.139
	SAT (Chen et al., 2022)	-	-	-	0.094
	GPS (Rampasek et al., 2022)	98.051	86.685	78.016	0.070
	Graph MLP-Mixer (He et al., 2023)	98.320	-	-	0.073
	GRIT (Ma et al., 2023)	98.108	87.196	80.026	0.059
GRED	GRED (Ding, Yuhui, et al., 2024)	98.383	86.759	78.495	0.077
	Ours	96.600	50.491	23.206	1.547

Experiments: Qualitative Analysis

Random Walk Strategy	Data Type	Dataset			
		MNIST Accuracy($\uparrow$)	PATTERN Accuracy($\uparrow$)	CLUSTER Accuracy($\uparrow$)	ZINC 12K MAE($\downarrow$)
Random Walk ($p = 1.0$, $q = 1.0$)	Integer	96.230	50.341	22.796	2.257
	Boolean	-	-	-	0.350
DFS-like Walk ($p = 1e10$, $q = 0.5$)	Integer	96.600	50.491	22.543	1.547
	Boolean	-	-	-	0.236
Highly DFS-like Walk ($p = 1e10$, $q = 0.1$)	Integer	96.600	49.704	23.206	1.547
	Boolean	96.840	52.242	25.887	0.250

Experiments: Ablation Study

Data Type	num_hops	Dataset			
		MNIST Accuracy($\uparrow$)	PATTERN Accuracy($\uparrow$)	CLUSTER Accuracy($\uparrow$)	ZINC 12K MAE($\downarrow$)
Integer	3	96.320	50.150	23.206	1.547
	4	96.420	50.531	22.670	1.547
	5	96.600	49.704	22.538	1.547
Boolean	3	96.890	52.153	25.887	0.241
	4	96.950	52.242	25.853	0.256
	5	96.840	52.132	25.831	0.236

Further Analysis

- Why we get lower performance than GRED?
 - **Fixed** and **not adaptive** random walk sequences
 - **Overfitting** problem & instability of the learning
- Why we fail to learn the model with PATTERN and CLUSTER dataset?
 - Node features are generated with a uniform **random** distribution (PATTERN)
 - Only one labeled node is **randomly assigned** to each community (CLUSTER)
- Why boolean type mask performs better than integer type mask?
 - Allow the model to learn **the structure information** of graphs (e.g. degree)

Ideas of Future Research

- Learnable p and q parameters in Random Walks
 - Enable **dynamic adjustment** of walk strategies
 - Implement **efficient optimization** to manage computational overhead
- Replace LRU with Mamba for Input-Dependent Parameters
 - Adapt parameters based on **node features** and **graph structure**
 - Enhance representation of **complex dependencies** within the graph
- Explore Effects of Boolean Masks in Random Walks
 - Improve feature extraction quality through **selective path filtering**
 - Mitigate over-smoothing and enhance **node distinguishability**

Conclusion

- Studied the limitations of various approaches in learning graph representations
- Proposed an approach of incorporating random walks and linear RNN to overcome such limitations
- Analyzed the effect of random walk strategies and provided future directions of applying random walks on graph representation learning

Q&A

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